

# Rajalakshmi Engineering College

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## 2024\_28\_III\_OOPS Using Java Lab

### 2028\_REC\_OOPS using Java\_Week 1\_MCQ

Attempt : 1  
Total Mark : 15  
Marks Obtained : 15

#### Section 1 : MCQ

1. Which of the following data types is used to store single characters?

**Answer**

char

**Status : Correct**

**Marks : 1/1**

2. What is the result of the following expression?

```
import java.util.*;
```

```
class ComplexExpressionExample {  
    public static void main(String[] args) {  
        int a = 5, b = 2, c = 3, d = 4;  
        int result = a + b * c / d - b;  
    }  
}
```

```
        System.out.println(result);
    }
}
```

**Answer**

4

**Status :** Correct

**Marks :** 1/1

3. What is the output of the following program?

```
class Arithmetic {
    public static void main(String[] args) {
        char ch = 'A';
        System.out.println(ch);
    }
}
```

**Answer**

A

**Status :** Correct

**Marks :** 1/1

4. What will be the output of the following code snippet?

```
import java.util.*;

class OperatorPrecedenceExample {
    public static void main(String[] args) {
        int a = 5, b = 3, c = 2;
        int result = a + b * c;

        System.out.println(result);
    }
}
```

**Answer**

11

**Status :** Correct

**Marks :** 1/1

5. What is the output of the following code?

```
class TestClass {  
    public static void main(String[] args) {  
        int a = 5;  
        int b = 10;  
  
        int sum = a + b;  
        int bitwiseAnd = a & b;  
        int bitwiseOr = a | b;  
  
        System.out.println(sum);  
        System.out.println(bitwiseAnd);  
        System.out.println(bitwiseOr);  
    }  
}
```

**Answer**

15015

**Status :** Correct

**Marks :** 1/1

6. What will be the output of the following program?

```
class DataTypesMCQ {  
    public static void main(String[] args) {  
        int a = 10;  
        double b = 5;  
        System.out.println(a / b);  
    }  
}
```

**Answer**

2.0

**Status :** Correct

**Marks :** 1/1

7. What will be the output of the following code snippet?

```
class DivisionExample {  
    public static void main(String[] args) {  
        double num1 = 10.5;  
        double num2 = 3;  
        int result = (int)(num1 / num2);  
        System.out.println(result);  
    }  
}
```

**Answer**

3

**Status :** Correct

**Marks :** 1/1

8. What is the output of the following code?

```
class TestClass {  
    public static void main(String[] args) {  
        int count = 8;  
        count = count ^ 1;  
  
        System.out.println(count);  
    }  
}
```

**Answer**

9

**Status :** Correct

**Marks :** 1/1

9. What is the output of the following code?

```
class TestClass {  
    public static void main(String[] args) {  
        int a = 10;  
        int b = 3;  
        System.out.println(a / b);  
    }  
}
```

**Answer**

3

**Status :** Correct

**Marks :** 1/1

10. What is the output of the following code?

```
class TestClass {  
    public static void main(String[] args) {  
        int x = 5;  
        int X = 10;  
  
        int sum = x + X;  
        int bitwiseResult = x | X;  
  
        System.out.println(sum);  
        System.out.println(bitwiseResult);  
    }  
}
```

**Answer**

1515

**Status :** Correct

**Marks :** 1/1

11. Which of the following data types is used to store floating-point numbers with greater precision?

**Answer**

double

**Status :** Correct

**Marks :** 1/1

12. Which of the following is not a primitive data type?

**Answer**

string

**Status :** Correct

**Marks :** 1/1

13. What will be the output of the following code?

```
import java.util.*;

class TernaryOperatorExample {
    public static void main(String[] args) {
        int a = 5, b = 10;
        int result = (a > b) ? a : b;
        System.out.println(result);
    }
}
```

**Answer**

10

**Status :** Correct

**Marks :** 1/1

14. What is the output of the following program?

```
class Demo {
    public static void main(String[] args) {
        String text = "Hello, World!";
        System.out.println(text);
    }
}
```

**Answer**

Hello, World!

**Status :** Correct

**Marks :** 1/1

15. What is the output of the following code?

```
import java.util.*;

class RelationalOperatorExample {
    public static void main(String[] args) {
        int x = 8, y = 4;
        boolean result = (x != y);

        System.out.println(result);
    }
}
```

**Answer**

true

**Status :** Correct

**Marks :** 1/1

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## 2024\_28\_III\_OOPS Using Java Lab

### 2028\_REC\_OOPS using Java\_Week 1\_Q1

Attempt : 1  
Total Mark : 10  
Marks Obtained : 10

#### Section 1 : Coding

##### 1. Problem Statement

Gloria is responsible for monitoring the performance of two machines in a factory. She needs to determine which of the two machines is operating closest to the optimal temperature of 100 degrees Celsius using the relational operator.

Assist Gloria in displaying the machine's temperature, which is closer to 100, and the difference from 100.

##### ***Input Format***

The first line of input consists of an integer N, representing the temperature of the first machine.

The second line consists of an integer M, representing the temperature of the second machine.

### **Output Format**

The output prints "The integer closer to 100 is X with a difference of Y" where X is the temperature of the closer machine and Y is the difference from 100.

Refer to the sample output for formatting specifications.

### **Sample Test Case**

Input: 90

80

Output: The integer closer to 100 is 90 with a difference of 10

### **Answer**

// You are using Java

```
import java.util.Scanner;
```

```
class Main{
```

```
    public static void main(String args[]){
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int n= sc.nextInt();
```

```
        int m= sc.nextInt();
```

```
        int a= Math.abs(100-n);
```

```
        int b= Math.abs(100-m);
```

```
        if(a<b){
```

```
            System.out.println("The integer closer to 100 is "+n+" with a difference of "+a);
```

```
        }else{
```

```
            System.out.println("The integer closer to 100 is "+m+" with a difference of "+b);
```

```
        }
```

```
    }
```

```
}
```

**Status : Correct**

**Marks : 10/10**

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## 2024\_28\_III\_OOPS Using Java Lab

### 2028\_REC\_OOPS using Java\_Week 1\_Q2

Attempt : 1  
Total Mark : 10  
Marks Obtained : 10

#### Section 1 : Coding

##### 1. PROBLEM STATEMENT:

Dave got two students who want help with their doubt. Each hands out an integer and wants to find if one Integer is Positive While the Other is Not Divisible by 3. Write a program to achieve this and conclude for them.

##### ***Input Format***

The first line of input represents the first integer.

The second line of input represents the second integer.

##### ***Output Format***

The output should display as "One of the integers is positive while the other is not divisible by 3." or "Neither of the integers meets the condition."

Refer to the sample output for the formatting specifications.

**Sample Test Case**

Input: 4

3

Output: One of the integers is positive while the other is not divisible by 3.

**Answer**

```
// You are using Java
import java.util.Scanner;
class Main{
    public static void main(String args[]){
        Scanner sc= new Scanner(System.in);
        int a= sc.nextInt();
        int b= sc.nextInt();
        if((a>0 && b%3!=0) || (b>0 && a%3!=0)){
            System.out.println("One of the integers is positive while the other is not
divisible by 3.");
        }
        else{
            System.out.println("Neither of the integers meets the condition.");
        }
    }
}
```

**Status :** Correct

**Marks :** 10/10

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## 2024\_28\_III\_OOPS Using Java Lab

### 2028\_REC\_OOPS using Java\_Week 1\_Q3

Attempt : 1  
Total Mark : 10  
Marks Obtained : 10

#### Section 1 : Coding

##### 1. Problem statement

Manoj, a developer at MoneyMatters Inc., is working on improving the company's financial system. He needs to create a program that takes an integer input, converts it into a double, and displays both the original integer and the converted double value.

##### ***Input Format***

The input consists of a single integer representing a monetary amount.

##### ***Output Format***

The first line of the output displays the "Original Integer: ", followed by an integer representation of the input value.

The second line displays the "Converted Double: ", followed by a double value representing the input as a decimal value.

Refer to the sample output for the formatting specifications.

**Sample Test Case**

Input: 20

Output: Original Integer: 20

Converted Double: 20.0

**Answer**

```
// You are using Java
import java.util.Scanner;
class num{
    public static void main(String args[]){
        Scanner sc= new Scanner(System.in);
        int a=sc.nextInt();
        System.out.println("Original Integer: "+a);
        double b=a;
        System.out.print("Converted Double: "+b);
    }
}
```

**Status :** Correct

**Marks : 10/10**

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## 2024\_28\_III\_OOPS Using Java Lab

### 2028\_REC\_OOPS using Java\_Week 1\_Q4

Attempt : 1  
Total Mark : 10  
Marks Obtained : 10

#### Section 1 : Coding

##### 1. Problem Statement

Vishal and Arun are discussing the properties of numbers. Vishal gives Arun two integers. He asks Arun to check if the sum of these two numbers is a multiple of their product.

Can you assist Arun and determine whether the sum is a multiple of the product?

##### ***Input Format***

The input consists of two space-separated integers.

##### ***Output Format***

The output prints:

1. "Sum is Multiple of Product" if the sum of the two numbers is divisible by their product.
2. "Sum is Not Multiple of Product" otherwise.

Refer to the sample output for formatting specifications.

**Sample Test Case**

Input: 1 2

Output: Sum is Not Multiple of Product

**Answer**

```
// You are using Java
import java.util.Scanner;
class Main{
    public static void main(String args[]){
        Scanner sc= new Scanner(System.in);
        int a= sc.nextInt();
        int b= sc.nextInt();
        int sum=a+b;
        int prod=a*b;
        if(sum%prod==0){
            System.out.println("Sum is Multiple of Product");
        }else{
            System.out.println("Sum is Not Multiple of Product");
        }
    }
}
```

**Status :** Correct

**Marks :** 10/10

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## 2024\_28\_III\_OOPS Using Java Lab

### 2028\_REC\_OOPS using Java\_Week 1\_Q5

Attempt : 1  
Total Mark : 10  
Marks Obtained : 10

#### Section 1 : Coding

##### 1. Problem Statement:

Emily has a beautiful circular garden in her backyard. She's interested in calculating two important measurements for her garden: the circumference and the area. To do this, she needs a program that can take the radius of her circular garden as input and provide the calculated circumference and area as output. The formulas she should use are as follows:

To calculate the circumference (C) of a circle, you can use the formula:

$$C = 2 * \pi * r$$

$$A = \pi * r^2$$

Where:

C represents the circumference.

A represents the area.

$\pi$  (pi) is approximately 3.14159.

r is the radius of the circle.

Emily is not a programmer, and she needs your help to create a program that will make these calculations for her garden.

### ***Input Format***

The first line of input contains a single double-point number radius, representing the radius of the circle.

### ***Output Format***

The output should consist of two lines:

The first line should print the circumference of the circle rounded to 2 decimal places, followed by the unit "meters".

The second line should print the area of the circle rounded to 2 decimal places, followed by the unit "square meters".

Refer to the sample output for formatting specifications.

### ***Sample Test Case***

Input: 3.0

Output: Circumference: 18.85 meters

Area: 28.27 square meters

### ***Answer***

```
// You are using Java
import java.util.Scanner;
class Main{
    public static void main(String args[]){
        Scanner sc= new Scanner(System.in);
        double r= sc.nextDouble();
        double c=2*3.14159*r;
```

```
double a=3.14159*r*r;  
System.out.println("Circumference: "+ String.format("%.2f", c) + " meters");  
System.out.println("Area: "+ String.format("%.2f",a) + " square meters");  
}  
}
```

**Status :** Correct

**Marks : 10/10**

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## 2024\_28\_III\_OOPS Using Java Lab

### 2028\_REC\_OOPS using Java\_Week 1\_Q6

Attempt : 1  
Total Mark : 10  
Marks Obtained : 10

#### Section 1 : Coding

##### 1. Problem Statement

Joey is learning about bitwise operations and is working on a project that involves extracting specific bits from integers. He needs to write a program that takes an integer and the number of bits N as input and outputs the value of the lowest N bits of the integer.

Help Joey in his project to understand and visualize how bitwise operations work in practical scenarios.

##### ***Input Format***

The first line of input consists of an integer X, representing the given integer.

The second line consists of an integer N, representing the number of bits to extract.

### **Output Format**

The output displays "Result: " followed by an integer representing the value of the lowest N bits of the given integer.

Refer to the sample output for formatting specifications.

### **Sample Test Case**

Input: 85

2

Output: Result: 1

### **Answer**

```
// You are using Java
import java.util.Scanner;
class Main{
    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        int x=sc.nextInt();
        int n=sc.nextInt();
        int a=(1<n)-1;
        int r=x&a;
        System.out.print("Result: "+r);
    }
}
```

**Status : Correct**

**Marks : 10/10**

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## 2024\_28\_III\_OOPS Using Java Lab

### 2028\_REC\_OOPS using Java\_Week 1\_Q7

Attempt : 1  
Total Mark : 10  
Marks Obtained : 10

#### Section 1 : Coding

##### 1. Problem Statement:

Miles is working on a program that involves analyzing two integers. He wants to check if either one of the integers is both:

Less than or equal to zero, and Odd. Can you help him create a program that identifies whether either of the integers meets these conditions?

##### ***Input Format***

The input consists of two integers on separate lines, denoted as 'input1' and 'input2'.

##### ***Output Format***

A single line with a boolean result (either 'true' or 'false') indicating whether either 'input1' or 'input2' is both less than or equal to zero and odd.

Refer to the sample output for format specifications

**Sample Test Case**

Input: -45

10

Output: true

**Answer**

```
// You are using Java
import java.util.Scanner;
class Main{
    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        int n= sc.nextInt();
        int m= sc.nextInt();
        if((n<=0 && n%2!=0) || (m<=0 && m%2!=0)){
            System.out.print("true");
        }else{
            System.out.print("false");
        }
    }
}
```

**Status :** Correct

**Marks : 10/10**

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## 2024\_28\_III\_OOPS Using Java Lab

### 2028\_REC\_OOPS using Java\_Week 1\_Q8

Attempt : 1  
Total Mark : 10  
Marks Obtained : 10

#### Section 1 : Coding

##### 1. Problem Statement

In the Kingdom of Finance, the royal treasury is managed by the treasurer, Sir Cedric. Sir Cedric tracks the daily expenses of the kingdom using an expense report that lists three major categories: food, clothing, and utilities. However, the King wants to know if the average daily expense is greater than at least two of these categories to ensure the kingdom is spending wisely.

Your task is to help Sir Cedric determine if the average daily expense is greater than two of the categories. Specifically, you need to calculate the average of the three expenses and check if it is greater than any two categories.

Note: Use the ternary operator

### ***Input Format***

Three integers a, b, and c represent the daily expenses for food, clothing, and utilities. Each integer is provided on a single line.

### ***Output Format***

The average of the three expenses, rounded to two decimal places.

A message indicating whether the average is greater than at least two of the expense categories.

1. If the average is greater than the two smallest monthly expenses, print "Average is greater than both X and Y," where X and Y are the two smallest expenses.
2. Otherwise, display "Average is not greater than two smallest expenses".

Refer to the sample output for formatting specifications.

### ***Sample Test Case***

Input: 4

6

10

Output: 6.67

Average is greater than both 4 and 6

### ***Answer***

```
// You are using Java
import java.util.Scanner;
class Main{
    public static void main(String args[]){
        Scanner sc= new Scanner(System.in);
        int a= sc.nextInt();
        int b= sc.nextInt();
        int c= sc.nextInt();
        double avg= (a+b+c)/3.0;
        System.out.printf("%.2f\n", avg);
        if(avg>a && avg>b){
            System.out.print("Average is greater than both "+a+ " and " +b);
        }
    }
}
```

```
else if(avg>b && avg>c){  
    System.out.print("Average is greater than both "+b+ " and " +c);  
}  
else if(avg>a && avg>c){  
    System.out.print("Average is greater than both "+a+ " and " +c);  
}  
else{  
    System.out.print("Average is not greater than two smallest expenses");  
}  
}  
}
```

**Status :** Correct

**Marks :** 10/10

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## 2024\_28\_III\_OOPS Using Java Lab

### 2028\_REC\_OOPS using Java\_Week 1\_Q9

Attempt : 1  
Total Mark : 10  
Marks Obtained : 10

#### Section 1 : Coding

##### 1. Problem Statement

Phill is a quality control manager at a manufacturing plant. He needs to verify if a sensor reading at a midpoint station (S2) falls exactly halfway between the readings of the previous station (S1) and the next station (S3). Help him by developing a program that checks if the second sensor reading is the average (midpoint) of the first and third sensor readings.

Use the relational operator to solve the program.

##### ***Input Format***

The first line of input consists of an integer S1, representing the sensor reading of the first station.

The second line consists of an integer S2, representing the sensor reading of the midpoint station.

The third line consists of an integer S3, representing the sensor reading of the next station.

### **Output Format**

The first line of output displays a boolean value representing whether the sensor reading at the midpoint station is halfway between the readings of the first and the next stations.

The second line displays one of the following:

1. If the result is true, print "The second integer is halfway between the first and third integers."
2. Otherwise, print "The second integer is not halfway between the first and third integers."

Refer to the sample output for formatting specifications.

### **Sample Test Case**

Input: 1

7

10

Output: false

The second integer is not halfway between the first and third integers.

### **Answer**

```
// You are using Java
import java.util.Scanner;
class Main{
    public static void main(String args[]){
        Scanner sc= new Scanner(System.in);
        int a= sc.nextInt();
        int b= sc.nextInt();
        int c= sc.nextInt();
        int ans= (a+c)/2;
        if(b==ans){
            System.out.println("true\nThe second integer is halfway between the first
and third integers.");
        }else{
```

```
System.out.println("false\nThe second integer is not halfway between the  
first and third integers.");
```

```
    }  
  }  
}
```

**Status :** Correct

**Marks :** 10/10

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## 2024\_28\_III\_OOPS Using Java Lab

### 2028\_REC\_OOPS using Java\_Week 1\_Q10

Attempt : 1  
Total Mark : 10  
Marks Obtained : 10

#### Section 1 : Coding

##### 1. Problem Statement

Aishu is supervising a construction project that needs to be completed with the help of three workers: A, B, and C.

She knows how many days each of them would take to complete the entire project individually:

A can complete it in x days, B in y days, C in z days.

Initially, all three workers (A, B, and C) work together for d1 days.

After that, C leaves, and only A and B continue for another d2 days.

Then B also leaves, and A works alone to finish the remaining work.

Your task is to help aishu to implement this functionality using the class WorkDistribution and Method calculateWork(int x, int y, int z, int d1, int d2)

Calculate the total work completed in the first  $d_1$  days by A, B, and C. Calculate the work completed in the next  $d_2$  days by A and B. Determine the remaining work after these  $d_1 + d_2$  days.

**Input Format**

The first line of input contains five space-separated integers:  $x$   $y$   $z$   $d_1$   $d_2$

where:

$x$  represents the Days A takes to complete the work alone

$y$  represents the Days B takes to complete the work alone

$z$  represents the Days C takes to complete the work alone

$d_1$  represents the Days A, B, and C work together

$d_2$  represents the Days A and B work together (after C leaves)

**Output Format**

The first line of output prints "Work done in first  $d_1$  days (A+B+C): " followed by a double value rounded to 2 decimal places.

The second line of output prints "Work done in next  $d_2$  days (A+B): " followed by a double value rounded to 2 decimal places.

The third line prints "Remaining work: " followed by a double value rounded to 2 decimal places.

Refer to the sample output for formatting specifications.

**Sample Test Case**

Input: 10 20 30 2 2

Output: Work done in first  $d_1$  days (A+B+C): 0.37

Work done in next  $d_2$  days (A+B): 0.30

Remaining work: 0.33

**Answer**

```
// You are using Java
import java.util.Scanner;
class Main{
    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        int x=sc.nextInt();
        int y=sc.nextInt();
        int z=sc.nextInt();
        int d1=sc.nextInt();
        int d2=sc.nextInt();
        double rateA=1.0/x;;
        double rateB=1.0/y;
        double rateC=1.0/z;
        double work1= (rateA+rateB+rateC)*d1;
        double work2= (rateA+rateB)*d2;
        double remain=1.0-(work1+work2);
        if(remain<0){
            remain=0.0;
        }
        System.out.printf("Work done in first d1 days (A+B+C): %.2f\n", work1);
        System.out.printf("Work done in next d2 days (A+B): %.2f\n", work2);
        System.out.printf("Remaining work: %.2f\n", remain);
    }
}
```

**Status :** Correct

**Marks : 10/10**